

Dimensionality reduction and unsupervised learning

Dataset: Olivetti faces, 400 photographs

How to use these notes

The primary text for this lecture is Géron, Chapters 7–8. These notes are not a replacement for it and do not repeat it: they are the lecture’s own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture’s notebook, and the course checks that mechanically rather than trusting it.

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1 The first lecture with no target column

Seven lectures, and every one was handed the answers: a price for every district, a digit for every image, a survivor flag for every passenger, a cover type for every plot. Today there is nothing. That changes what a metric can be, what a baseline is, and how you find out you were wrong.

Supervised	Unsupervised
You are given $(\mathbf{x}^{(i)}, y^{(i)})$	you are given $\mathbf{x}^{(i)}$ only
The metric compares prediction to truth	the metric compares the grouping to itself
The baseline is “predict the majority”	the baseline is “group at random”
You know when you are wrong	you find out when someone looks

The consequence for the whole hour

Every number computed today is a statement about *geometry*, not about people. Whether the geometry is the people is a separate question, and it is answered by looking.

1.1 The brief

“We have a photographic archive. We do not know how many distinct people are in it. Group the photographs by person, so that a human can name a group instead of naming every photograph.” Nobody remembers who is in it, an annotator costs money per photograph, and we are funded for *forty* labels. That is the ordinary situation outside a benchmark: data is cheap, labels are not.

400 photographs of 40 people, ten each, 64×64 pixels flattened to $\mathbf{x}^{(i)} \in \mathbb{R}^{4096}$ with $x_j \in [0, 1]$ — 6.6 MB for the whole corpus. Ten times more dimensions than points.

The rule that keeps this honest

Olivetti ships with the identity of every photograph; the archive we are pretending to hold does not. So y is used in exactly two ways, both announced where they happen: as the *forty labels we paid for*, and as an *audit*, to find out what those forty could not have told us. Any other use of y is the same error as scoring on the test set, wearing different clothes.

1.2 Before anything is fitted: can this work at all?

k -means groups points close in Euclidean distance, so one measurement decides whether the hour is worth spending. It is an audit — it uses the hidden labels.

Euclidean distance, raw pixels	Pairs	Mean	Median
Same person	1,800	8.38	8.15
Different people	78,000	12.41	12.15

The gap is real and it is small: 6% of same-person pairs sit further apart than the *median* different-person pair.

Failure condition — stated in advance

No method that groups by Euclidean distance in raw pixels will recover the forty people cleanly. Not because the method is bad — because the representation puts a lamp and a nose on the same axis, with the same weight.

Moving the light source changes hundreds of pixel values by a large amount. Changing the person changes a few dozen by a small one. Euclidean distance adds them all up equally. Measuring this costs one cell and tells you whether you are about to tune or about to hope.

It is also not free to work there. One model-selection sweep — k -means at every k from 2 to 60, ten restarts each, scored by silhouette — costs 77 seconds for 590 runs of Lloyd’s algorithm at $O(\text{iterations} \cdot mkn)$ and 2 more for 59 silhouette scores at $O(m^2n)$. Both cost models are linear in n . Two of the three factors are properties of the problem; the third is a property of the *representation*.

2 Derivation: PCA via the SVD

Two questions. Among all d -dimensional subspaces, which loses the least when you project onto it, and how much does it lose? And if all you need is that *distances* survive, how small can d be?

Write $\mathbf{X} \in \mathbb{R}^{m \times n}$ with one photograph per row, $\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^\top$ the centred data, $\mathbf{W} \in \mathbb{R}^{n \times d}$ with $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_d$, and $\mathbf{P} = \mathbf{W}\mathbf{W}^\top$ the orthogonal projector.

Failure condition — an SVD taken without centring

Without centring, the direction of largest $\sum_i \|\mathbf{w}^\top \mathbf{x}^{(i)}\|^2$ is essentially $\bar{\mathbf{x}}$ itself, and the first component describes nothing but “this is a photograph of a face”.

Scikit-Learn’s PCA centres for you. If you write the SVD by hand it does not, and the result is a plausible-looking first component and no error message. It is the commonest silent bug in a hand-rolled PCA.

Seven steps

1. The objective. Project each centred face and measure how far it had to move:

$$E(\mathbf{W}) = \sum_i \|\tilde{\mathbf{x}}^{(i)} - \mathbf{W}\mathbf{W}^\top \tilde{\mathbf{x}}^{(i)}\|_2^2 = \|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{W}\mathbf{W}^\top\|_F^2.$$

2. Pythagoras. $\mathbf{P}$ is an orthogonal projector, so kept and lost are orthogonal, and since $\|\tilde{\mathbf{X}}\|_F^2$ does not depend on $\mathbf{W}$,

$$E(\mathbf{W}) = \|\tilde{\mathbf{X}}\|_F^2 - \|\tilde{\mathbf{X}}\mathbf{W}\|_F^2.$$

3. As a trace. $\|\tilde{\mathbf{X}}\mathbf{W}\|_F^2 = \text{tr}(\mathbf{W}^\top \mathbf{C} \mathbf{W})$ with $\mathbf{C} = \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$.

4. The SVD diagonalises it. With $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$,

$$\mathbf{C} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top,$$

so the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{C}$ with eigenvalues σ_j^2 . They are the principal components.

5. Rotate the unknown. Write $\mathbf{W} = \mathbf{V}\mathbf{A}$ with $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_d$; then $\text{tr}(\mathbf{W}^\top \mathbf{C} \mathbf{W}) = \sum_j \sigma_j^2 h_j$, where h_j is the squared norm of row j of $\mathbf{A}$, constrained to $0 \leq h_j \leq 1$ with $\sum_j h_j = d$.

6. A weighted sum with d units to spend. The σ_j^2 are sorted, so put a full unit on each of the d largest and nothing on the rest: the maximum is $\sum_{j \leq d} \sigma_j^2$, attained at $\mathbf{W} = \mathbf{V}_d$.

7. The error that is left. Since $\|\tilde{\mathbf{X}}\|_F^2 = \text{tr}(\mathbf{C}) = \sum_j \sigma_j^2$,

$$\min_{\mathbf{W}} E(\mathbf{W}) = \sum_{j > d} \sigma_j^2.$$

That is Eckart–Young–Mirsky in the Frobenius norm. “PCA minimises reconstruction error” is a *theorem*, not a slogan — and it names the price in advance: the tail of the spectrum, readable before you choose d .

Note also that step 2 says minimising error and maximising projected variance are *the same problem*, not two related ones. And $\mathbf{C}$ is the matrix Lecture 2’s normal equation had to invert, so step 4 says exactly when it cannot be: when some $\sigma_j = 0$.

A theorem you can check to machine precision

```
lhs = ((X_centred - Xd) ** 2).sum() # the rank-d truncation's error
rhs = (S[d:] ** 2).sum()           # the tail of the spectrum
assert abs(lhs - rhs) / rhs < 1e-5
```

Two sides computed by entirely different routes. When a theorem hands you an identity, *assert the identity*: it is the cheapest possible test that your code is the theorem.

2.1 Choosing d without guessing

By step 7 the fraction of squared error removed by keeping d components is exactly $\sum_{j \leq d} r_j$, where $r_j = \sigma_j^2 / \sum_i \sigma_i^2$. The explained variance ratio is not a heuristic; it is the theorem, normalised.

The first component alone holds 23.8%, ten hold 65.6%, fifty hold 87.4%. Cumulating to a target: 66 components reach 90%, **123 reach 95%**, 260 reach 99%. The rest of this lecture runs at 123 — a factor of 33 fewer numbers per photograph.

A principal component is a vector in $\mathbb{R}^{4096}$, so reshaping it to 64×64 makes it an image; on faces these are called **eigenfaces**. That is not a property of faces but of PCA: the components are directions in the input space, so whatever an input looks like, a component looks like too. And looking at the first few, they are *lighting* — exactly what the distance histogram predicted would dominate.

2.2 The second question: what if you only need distances?

PCA looks at the data and spends an SVD finding the best subspace. The alternative is to not look at the data at all: draw $\mathbf{R} \in \mathbb{R}^{n \times d}$ with independent Gaussian entries and set $\mathbf{Z} = \frac{1}{\sqrt{d}} \mathbf{X} \mathbf{R}$.

The Johnson–Lindenstrauss lemma

For any $0 < \varepsilon < 1$ and any set of m points in $\mathbb{R}^n$, there is a linear map f into $\mathbb{R}^d$ with

$$d \geq \frac{4 \log m}{\varepsilon^2/2 - \varepsilon^3/3}$$

such that for every pair $\mathbf{u}, \mathbf{v}$,

$$(1 - \varepsilon) \|\mathbf{u} - \mathbf{v}\|^2 \leq \|f(\mathbf{u}) - f(\mathbf{v})\|^2 \leq (1 + \varepsilon) \|\mathbf{u} - \mathbf{v}\|^2.$$

The tolerance is on *squared* distances. Quoting ε against unsquared ones understates the distortion by about a factor of two.

Read the formula again. What is missing?

In it: m , the number of points, logarithmically; and ε , the tolerance. *Not* in it: n , the dimension you are starting from.

400 points need the same target dimension whether they live in 4,096 dimensions or in four million.

ε	400 points	One million points
0.1	5,135	11,841
0.2	1,382	3,188
0.3	665	1,535
0.5	287	663

Two and a half thousand times as many points costs about twice the dimension. That is the $\log m$, and it is why the theorem is interesting at all.

But be honest about the top-left cell: 5,135 dimensions to hold every pairwise distance among 400 faces to within 10% is *more than the 4,096 we started with*. The bound is a worst case over all point sets — it has to work for the nastiest configuration anyone can construct, and a photographic archive is not that. So it is sufficient, never necessary, and the reason to teach it is not the constant. Measured on our own distances at the bound's own $d = 1,382$, the worst of the 79,800 pairs is distorted by 0.17: inside the 0.2 promised, and not by much.

3 Reducing the dimension

Method	What it does	Fit time, s	Held-out error
PCA, full SVD	the complete SVD, then truncate	0.074	0.00240
PCA, randomised	approximates the top d	0.081	0.00241
Incremental PCA	one batch at a time	0.142	0.00240
Random projection	never looks at the data	0.006	0.32093

Three PCA variants land on the same error at different prices. Random projection is essentially free and pays with a reconstruction error 133 times larger — a random 123-dimensional subspace of $\mathbb{R}^{4096}$ keeps only about $123/4096 = 3.0\%$ of a generic vector's squared norm.

That is not a bug in random projection

PCA minimises exactly this column by construction. Johnson–Lindenstrauss promises something else entirely — that *relative* distances survive. Neither theorem is about the other quantity.

So the question is never which is better. It is whether you need the reconstruction or only the distances.

PCA is the wrong tool under three conditions. *Variance is not signal*: if the largest variance is a nuisance — a lamp — PCA keeps the nuisance first. *The structure is not linear*: a subspace is flat. *The features are not commensurable*: PCA is not scale-free, so standardise first unless, as here, every feature is already the same kind of number. The first condition is the one on this data, and no amount of compression repairs it.

3.1 One procedural condition, because it is invisible

A reducer is an estimator: it is *fitted*. Fit it before the split and the held-out faces have helped choose the subspace they are then judged in. The 280 training faces span at most a 280-dimensional subspace of $\mathbb{R}^{4096}$, so adding 120 more points genuinely changes which directions survive.

$d = 123$, 20 stratified splits	Fitted on train	Fitted on everything	Leaky wins
Reconstruction error, held out	0.00243	0.00096	20 of 20
Accuracy, held out	97.4%	97.4%	3 of 20

Failure condition — the decision rule, and why it is still procedural

The leak makes the reconstruction error look 61% better and moves the accuracy by nothing measurable.

Fitting an unsupervised step on everything costs you exactly when *that step's own objective is the number you report*. Reconstruction error is that objective; downstream accuracy is not — and you cannot tell which row you are in without doing the split. So put it in a Pipeline, which refits inside every fold and makes the leak structurally impossible rather than merely avoided.

4 Clustering, and how to choose k

k -means makes four assumptions whether you asked or not: the partition is a Voronoi diagram, so every cluster is *convex*; the only quantity is distance to a centre, so clusters are *spherical and equal-sized*; every point belongs to exactly one cluster, so there is no “none of these”; and k is *given*. Three of the four are false here.

Metric	Needs	Available here?
RMSE, accuracy, F_1 , ROC AUC	a truth per instance	no
Inertia	nothing	yes
Silhouette	nothing	yes
Adjusted Rand index	labels — a subset is enough	on the forty

Two of these can choose a model. One can be shown to a stakeholder. They are not the same one, and keeping those jobs apart is this course's standing rule applied where there is no test set at all.

4.1 Why inertia cannot choose k

$$J(k) = \sum_i \|\mathbf{x}^{(i)} - \boldsymbol{\mu}_{c(i)}\|_2^2$$

is what k -means minimises — and it is non-increasing in k , with $J(m) = 0$ exactly. Refining a partition cannot increase the sum of squared distances to the assigned centre. *There is no interior minimum to find, at any k , on any dataset*, so “minimise inertia” selects one cluster per photograph. The elbow rule exists to work around that, and on our curve the kneedle rule picks $k = 19$ where the archive has 40 — with 46% of the inertia at $k = 2$ still present at $k = 40$.

The elbow is not a criterion

It is a picture of a criterion that does not exist, and no better heuristic repairs it, because the information is not in J .

The silhouette does have an interior maximum. For a point i , with a its mean distance to its own cluster and b to the nearest other,

$$s_i = \frac{b - a}{\max(a, b)} \in [-1, 1],$$

penalised both for splitting a group and for merging two. It costs $O(m^2n)$ — every pair, over every feature, and both factors matter.

4.2 Before reading any silhouette: what does nothing score?

Random assignment	Silhouette	ARI on the forty
$k = 10$	-0.0422 ± 0.0045	-0.0040 ± 0.0205
$k = 40$	-0.1288 ± 0.0107	-0.0089 ± 0.0244
$k = 60$	-0.1805 ± 0.0124	-0.0075 ± 0.0261

Both anchors sit at zero, and now you know the width of the noise around zero. A silhouette of 0.02 is not a discovery.

The **adjusted Rand index** compares two groupings pair by pair and subtracts what chance would give, so random scores 0 in expectation at any k and perfect scores 1. We may compute it on the forty labelled photographs only — 780 pairs, of which just 22 are the same person — so treat it as a sanity check with a wide interval, never as a ranking.

	k	Silhouette	ARI on the forty
Random assignment	40	-0.1288	-0.0089
The silhouette's choice	59	0.1707	0.3954
The true k — an audit	40	0.1465	0.2978

Well above the anchor, so the clustering is doing something. It chose 59 groups for 40 people, and the silhouette at the truth is *lower* than at its own choice.

4.3 The mean hides the shape

A silhouette diagram draws one knife per cluster, width for size and length for quality. A knife entirely left of the mean marks a cluster worse than average; knives of very unequal width mean one group has swallowed several; a knife extending into negative s_i marks points closer to another cluster than their own; and *every knife short* means the clusters overlap and the geometry has no clean gaps. Ours is the last, at every k we tried — which is the diagnosis the mean of 0.1707 does not contain.

At $k = 40$, against the hidden identities	Value
Cleanest cluster: 10 photographs, purity	1.00
Worst cluster: 24 photographs of 10 people, purity	0.21
Mean purity across the forty clusters	0.74
Clusters holding exactly one person	16
Clusters with exactly ten members	7
Cluster sizes, smallest to largest	4 to 25

Every true group has exactly ten members. The failure condition stated at the start of the lecture is now a measurement — and the worst cluster is not a person at all, but a lighting condition: several different people lit and posed the same way.

4.4 The score that chose the model also reports it

Failure condition — a bias that needs no labels to exist

Sweep k , keep the best silhouette, print it, and the number that chose the model is the number that reports it. Measured: choose k on a random half of the corpus and score the chosen model on the other half, twenty times.

Silhouette, 20 random halves	Mean	sd
Reported — chose and scored on the same half	0.1777	0.0123
Held out — had no vote	0.0878	0.0154
Optimism	0.0899	0.0229

Half the reported score, and worse in 20 splits out of 20.

5 Putting the two halves together

The clustering code is unchanged — same grid, same seed, same `n_init`. Only what k -means is looking at changed.

Dimensions	Sweep, s	Best k	Silhouette	ARI, all 400
4,096 — raw pixels	78	59	0.171	0.537
20	1.9	58	0.286	0.540
50	1.7	59	0.256	0.493
100	1.9	60	0.218	0.547
123 — 95% of the variance	2.1	60	0.205	0.524
399	5.8	59	0.171	0.537

The sweep is 37 times faster at 123 dimensions and the agreement with the true identities does not fall. The ARI column is there because a speed-up that quietly costs quality is not a speed-up. The seconds are a property of one machine at two threads; the *shape* — an order-of-magnitude saving at no cost in ARI — is a property of the method.

Failure condition — careful: the silhouette went up

That is not evidence the clustering improved. The silhouette is computed in whatever space you hand it, and compressing the data changes every distance, so the criterion is *not comparable across the rows of that table*.

Which is exactly why the ARI column is there: it is measured against the identities and does not depend on the representation. Never compare two scores measured on different evaluation sets — and a different representation is a different evaluation set.

6 Beyond k -means

Compression fixed the clock. It did not fix the geometry.

	DBSCAN	Gaussian mixture
Cluster shape	any connected dense region	an ellipsoid
Number of clusters	found, not given	given
Every point assigned?	no — there is a noise label	softly, with a probability
Main parameter	eps, a distance	n_components

Reporting a method that did not work

DBSCAN needs one density scale that separates clusters from gaps. Best over the whole grid: ARI 0.133 at $\text{eps}=6.0$, with 41 clusters and 97 of the 400 faces called noise; it never finds more than 50 clusters and never 40.

Forty groups of ten faces, all lit by the same rig, have no such scale — the gap between two people is about the gap between two photographs of one person. This is a real result and it belongs in the report. A method that fails for a stateable reason is more informative than a method that was never tried.

A Gaussian mixture models the data as k Gaussians fitted by expectation–maximisation, so clusters may be elongated and of different sizes, assignment is soft, and — being a density model — it also scores how likely a new point is.

Why this needed the first half of the lecture

A full covariance in n dimensions has $n(n + 1)/2$ free parameters.

Setting	Free parameters
One full covariance in 4,096 dimensions	8,390,656
Forty of them	335,626,240
Forty diagonal components in 123 dimensions — the whole model	9,880

Estimated from 400 photographs. The first is not slow — it is *undefined*: every covariance estimate is singular. Arithmetic settles that in one line, where an afternoon of waiting does not.

BIC picks 5 components while the ARI peaks at 55. Both penalise complexity, but they answer a different question: BIC asks which model best explains this data *as a density*, and we are asking which grouping matches the people. Those coincide only when the people really are Gaussian blobs, and here they are not.

6.1 Anomaly detection, two ways

Plant twelve corrupted images — four rotated, four dimmed, four double-exposed — and count how many each detector puts in its top twelve. Reconstruction error caught **12 of 12**; the mixture density caught 6.

A rotated or double-exposed face is outside the subspace the eigenfaces span, so its reconstruction

error is large: median 0.0017 against 0.0002 for a genuine face. The mixture density is estimated *inside* that subspace, so a corruption projecting onto an ordinary-looking point is invisible to it. Use reconstruction error when the anomaly is a different *kind* of object; use density when it is an ordinary object in an unusual *place*.

7 Spending the forty labels

Where they are spent turns out to matter more than what is done with them. Same classifier, same held-out faces, twenty splits; the only thing that varies is which forty were labelled.

Labels used	Held-out accuracy	sd over 20 splits
40 at random	48.1%	3.8%
40, one per cluster	60.5%	4.9%
Propagated to the whole cluster	55.7%	4.7%
Propagated to the closest 75%	55.1%	4.8%
All 280 true labels — the ceiling	97.4%	1.6%

Twenty splits, because the differences between the middle three rows are of the same order as their spread. *Propagation buys 280 labels from 40 and does worse than the 40 alone*, because a wrong propagated label is worse than no label. Report that; do not bury it.

7.1 The diagnosis was right, and the obvious repair still fails

Representation	Best k	Silhouette	ARI, all 400
PCA to 123 dimensions	60	0.205	0.524
PCA 123, first 3 components discarded	60	0.184	0.508
Unit-norm each image, then PCA 123	58	0.189	0.464

Neither helps. The leading components carry pose and face shape as well as illumination, so removing them throws away signal with the nuisance. The remedy that works is a representation learned *for the task* rather than for variance — which is what Part VI builds, and why a pretrained encoder beats a hand-chosen projection.

Note also what chose k there: sweeping k and keeping the best ARI would make both repairs look like gains, because ARI is computed from the labels this whole lecture is pretending not to have.

8 The notebook

What to change

Change `n_components=0.95` to `0.99`, which takes d from 123 to 260. The sweep slows by roughly the ratio of the two and the ARI barely moves — the whole trade-off in one edit.

9 Exercises

1. PCA is derived by maximising $\|XW\|_F^2$ subject to $W^T W = I$. State why the data must be centred first, and what the first component becomes if it is not. [5]
2. The Johnson–Lindenstrauss bound, evaluated for 400 images at $\varepsilon = 0.1$, asks for more dimensions than the data has. What do you conclude about the bound, and about the method? [5]
3. A pipeline fits PCA on the whole dataset and then splits. Name what leaked, and say whether the leak grows or shrinks as the corpus grows. [4]
4. Inertia always falls as k rises, so it cannot choose k . State what the silhouette adds, and one situation where it also fails. [5]
5. You have 4,096-dimensional face vectors and are told two faces are “close”. Explain why that alone does not mean they are the same person. [4]

Summary

PCA is the SVD of the centred data: the principal subspace minimises squared reconstruction error exactly, and the error it leaves is $\sum_{j>d} \sigma_j^2$, readable before you choose d . Minimising that error and maximising the projected variance are the same problem by Pythagoras, and the explained variance ratio is that theorem normalised — 123 components hold 95% of 4,096 pixels. Johnson–Lindenstrauss bounds the target dimension by the number of points and the tolerance, never by the dimension you started in, and it is a loose worst case, so measure the distortion you actually get.

Inertia cannot choose k : it is monotone in k and zero at $k = m$. The silhouette can, at $O(m^2n)$, must be read against a random anchor, and is optimistic by about half when it both chooses and reports. Compressing to 123 dimensions made the sweep 37 times faster at no cost in ARI — but the silhouette rose, and that number is not comparable across representations.

The lecture opened by measuring that pixel distance is partly identity and partly lighting, and closed by confirming it: mean cluster purity 0.74, the worst cluster a lighting condition rather than a person, DBSCAN failing for a stateable reason, and two one-line repairs that both made things worse. And forty labels spent on cluster representatives were worth a quarter more accuracy than forty spent at random — while propagating them to 280 was worse than not propagating at all.